

CPP NOTES – DAY 36

Constexpr

The constexpr specifier, introduced in C++11, allows the evaluation of functions and variables at compile time, enhancing performance by reducing runtime computations. This feature is particularly useful for optimizing programs by performing calculations during **compilation** rather than execution.

Association, Aggregation and Composition

Relationship	Ownership	Lifetime Dependency	Example
Association	No	Independent	Teacher ↔ Student
Aggregation	Shared (weak)	Independent	Team ↔ Player
Composition	Exclusive (strong)	Dependent (part dies with whole)	House ↔ Room

FYI:

- *Can I call function at compile time? Yes using constexpr*
- *In constexpr functions, **stack frame will not be generated**, it will directly executed*
- *Association – do not have the control of one object in another, hence it is achieved by pass by value.*
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